

MALWARE HANDLING PROTOCOL

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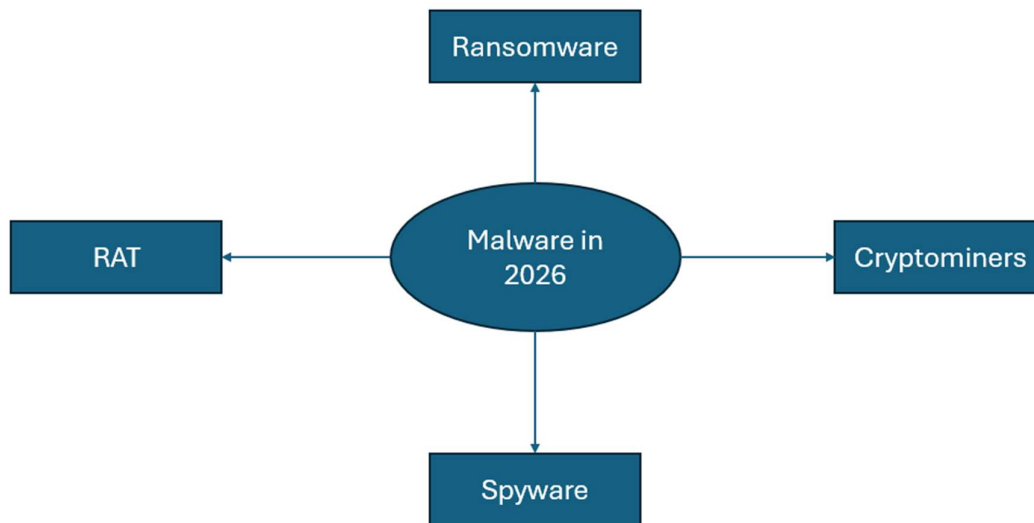
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Introduction

The malware infection handling protocol is designed to aid users and companies in removing malware in the case that malware has managed to infiltrate your device. This protocol was developed as sometimes malware can bypass defences meant to prevent it from infiltrating systems. This malware infection protocol will be mainly for Windows Operating systems. However, there will also be an example of Linux for those who may be on Linux operating systems. There are currently 5 main types of malware used by attackers in 2026. These include Ransomware, RATs (Remote Access Trojans), Spyware, Cryptominers and Loaders (Used to deliver other types of malware).

This malware infection handling protocol may not always be able to remove malware successfully from a device. This is because sometimes more sophisticated malware can avoid AV detection or infect firmware-level processes of the device. However, this protocol can heavily reduce and provide preparedness if malware were to infiltrate your device. Do not blindly trust this protocol as your only way to detect and remove malware. Make sure to adapt this protocol to the types of malware your threat model normally uses.



The spider diagram above shows the 4 most common types of malware used by attackers in 2026. This spider diagram does not include nation-state level malware. It also excludes loaders as they are used for delivering malware instead of causing direct harm to devices or data. Look at the paragraph above for more details on loaders and do your own research.

Main Malware Handling Protocol

- This section was made to remove the most common types of malware and should work well for the average user.
- This section will not be effective for nation-state level malware or ransomware.
- The first step in this protocol is to calm down
- This can be done by taking a few deep breaths or quickly releasing anger in a non-harmful way
- This step is important to reduce emotional noise, and it allows for more focus in removing malware.
- The second step is to establish a performance baseline for your device
- This step means having known verified baseline for how your device performs when Idle
- This baseline includes CPU usage, Disk activity, Network usage and RAM usage
- This can be done by using task manager on Windows or On Linux, you can use commands such as htop or Glances.
- This step can be skipped if you have already established a baseline
- This step is important to be able to detect malware as malware usually impacts the performance of your device or system.
- The third step is detection
- This means you will be detecting the malware on your device
- This can be done by noticing signs like for example slower performance or a malicious task in task scheduler
- You can cross-reference the current device performance against the baseline by closing all applications and non-essential processes.
- This step is important to confirm the presence of malware and prevent false positives at times
- The fourth step is containment
- This step means containing the spread of the malware
- This can be done by disconnecting the device from the network and changing passwords of accounts connected to the PC by using a separate clean device.
- This step is important because it reduces the blast radius heavily
- The fifth step is removal
- This step means removing the malware which can be done by using AV
- For example, you may use Windows Defender full scan or custom scan if you know where the malware is exactly located.
- This step is important to make sure that the malware is no longer affecting your device
- The final step is documenting
- This can be done through the structure What (Type of malware) → How (How did it access your device) → Patch(How did you remove it) → Improvements (How could I improve for next time?)

Malware Removal Failure Section

- This section was made for the case where malware removal is not successful from the initial steps.
- It was also designed for nation-state level malware and ransomware removal.
- Users with threat models such as organised cybercriminals or nation-state actors are the most likely to need this section.
- The first step is to calm down once again
- This is because unsuccessful malware removal can be quite stressful
- This step can be done with deep breaths or releasing anger in a non-harmful way if needed.
- This step is important as malware removal failure or sophisticated malware can be extremely stressful to deal with.
- The second step is advanced removal
- This is because sometimes malware can be in the Firmware or Kernel processes of the device
- This step can be done through Windows Defender's offline scan or using Linux-supported AV with kernel scanning capabilities.
- This step is important as it can increase the chances of removing sophisticated malware.
- The third step is recovery
- This means restoring the data or operating system on the device via the usage of backups
- This can be done through clean installation of the operating system via USB or full system backups
- Other data like files and folders can be downloaded from cloud storage or physical drives
- This step is important as ransomware even after removal requires recovery to restore data.
- The final step is documentation
- This means documenting the process of the more advanced removal and recovery.
- This can be done through the structure What (What was the malware?) → How (How did you remove the malware?) → Recovery (What was done for recovery?) → Improvements (What could be improved for next time?)
- This step is important as it allows you to learn from the attempt and improve your incident response skills.
- Make sure to adapt the steps and tools based on your operating system and current resources.

Malware Handling Protocol Workflow

- This workflow will show an example of a user dealing with a RAT and then ransomware using the protocol.
- First, the user notices unusual behaviour on their device with their mouse moving randomly and applications being open.
- Second, the user takes a deep breath as it can be quite stressful and annoying to deal with a RAT on a device.
- Third, the user confirms the RAT on their device by checking their current device's performance against baseline.
- The user finds that their CPU, memory, and network usage is higher than usual while the device is idle.
- Fourth, the user immediately disconnects their device from the internet to cut off the RAT's connection to their device.
- The user using their phone changes the passwords of all accounts that were connected to their laptop.
- Fifth, the user runs a full scan of their device using windows defender.
- Windows defender finds the RAT and successfully removes it from the device
- Sixth, the user documents the RAT malware infection.
- The user writes "The type of the malware was a remote access tool which got onto my device as I downloaded cracked software. I removed the RAT by running a full scan in windows defender. Next time I will make sure to avoid downloading cracked software."
- 3 months later, the user encounters ransomware on their device.
- First, the user gets a ransom note on their screen asking them to pay a ransom price in cryptocurrency.
- Second, the user takes a few deep breaths as ransomware can be stressful and increasingly frustrating to handle.
- Third, the user runs a full scan with windows defender.
- Windows defender locates the ransomware encryption process and removes it successfully from the device.
- Fourth, the user restores the system via USB completely reinstalling the operating system.
- The user loses all the data on their device and must rely on backups via cloud, physical drives and other devices.
- Fifth, the user documents the ransomware attempt.
- The user writes "The type of malware was ransomware which may have gotten on my device through SMB or another vulnerability in a network protocol I used. I have not fully confirmed the cause. I removed the ransomware by running a full scan on windows defender and recovered by reinstalling the operating system. I will make sure to implement controlled folder access and lock down ransomware entry points"
- The user continues using their device as normal after recovering their data.

Malware Handling Protocol Trade-offs

- **This malware handling protocol has trade-offs just like any other system I have built so far.**
- **Users may face further implementation challenges depending on their resources and operating systems.**
- **I have identified 4 main trade-offs with this protocol.**
- **The first trade-off is cognitive load.**
- **This trade-off exists because it can be mentally/emotionally expensive to establish a baseline, follow the steps of the protocol and document the attempt.**
- **This means that the user is more likely to make mistakes or avoid following the protocol if its too high of a cognitive load.**
- **This trade-off can be managed by minimising the steps to what you can remember and ensuring that real-time protection is on.**
- **The second trade-off is removal failure.**
- **This trade-off exists as sometimes standard removal may fail especially when dealing with sophisticated malware.**
- **This means that some users may struggle or experience further compromise if standard removal were to fail.**
- **This trade-off can be managed by using the removal failure section which includes backups, kernel-level scanning and more advanced methods.**
- **The third trade-off is data loss.**
- **This trade-off exists as certain types of malware like ransomware can cause the need for recovery.**
- **This means that data loss occurs as the data on the device is corrupted by the malware infection.**
- **This trade-off can be managed by creating backups, storing data in multiple locations and preparing to accept some data loss.**
- **The fourth trade-off is cost.**
- **This trade-off exists as some of the best anti-virus software can be extremely expensive.**
- **This means that users may struggle to have kernel/boot level scanning capabilities if they are operating on limited budget.**
- **This trade-off can be managed by using built-in tools, and behavioural habits such as avoiding cracked software to avoid malware infecting your device in the first place.**